

Problem 1

Let X be an infinite set and let $B(X)$ be the space of all bounded functions $X \rightarrow \mathbb{R}$, equipped with the norm $\|f\| = \sup_{x \in X} |f(x)|$ and the corresponding metric. Show that the metric space $B(X)$ is not separable.

Solution. Assume the contrary that there exists a countable dense subset of $B(X)$, $D = \{d_i\}_{i=1}^{\infty}$. Consider $2^X = \text{Hom}(X, \{0, 1\}) \subseteq B(X)$. Note that for $f, g \in 2^X$ where $f \neq g$ we have that

$$\|f - g\|_{\infty} = 1$$

Thus

$$B_{\frac{1}{3}}(f) \cap B_{\frac{1}{3}}(g) = \emptyset$$

Since $\{d_i\}_{i=1}^{\infty}$ is dense, $\forall f \in 2^X$ there exists $d_{k_f} \in B_{\frac{1}{3}}(f) \cap D$. Since the open balls are all mutually disjoint, each open ball $B_{\frac{1}{3}}(f_i)$ maps to a unique d_{k_i} . Let $P : 2^X \rightarrow D$, where $f \rightarrow d_{k_f}$. The mapping is naturally injective. Thus we have an injective mapping from an uncountable set to a countable one. Hence we have a contradiction, and $B(X)$ is not separable.

Problem 2 : Basic Exam, Spring 2019

Show that each metric space can be embedded isometrically into a Banach space: in other words, given a metric space (X, d) , show that there exist a Banach space (i.e. a complete normed space) $(V, \|\cdot\|)$ and a map $\Phi : X \rightarrow V$ such that

$$\|\Phi(x) - \Phi(x')\| = d(x, x'), \quad \forall x, x' \in X.$$

Solution. $(C(X), \|\cdot\|_{\infty})$ is a Banach Space of bounded continuous functions from X to $\mathbb{R}$. Fix a $x_0 \in X$. Let $\Phi : X \rightarrow C(X)$ be the following map $x \rightarrow [y \rightarrow d(x, y) - d(x_0, y)]$.

1. Continuity of $\Phi(x)$ for all $x \in X$: $\forall \varepsilon > 0$, $\forall y, y' \in X$ such that $d(y, y') < \varepsilon$ we have that:

$$|\Phi(x)(y) - \Phi(x)(y')| = |d(x, y) - d(x, y')| \leq d(y, y') < \varepsilon$$

2. Boundedness of $\Phi(x)$ for all $x \in X$: $\forall y \in X$:

$$\begin{aligned} |\Phi(x)(y)| &= |d(x, y) - d(x_0, y)| \\ &\leq d(x, x_0) \end{aligned}$$

Thus we have that

$$\|\Phi(x)\|_{\infty} \leq d(x, x_0) < \infty.$$

Thus $\Phi(x)$ is a well defined map. We will show that it's an isometry, and an embedding. $\forall x, y, z \in X$:

$$|\Phi(x)(z) - \Phi(y)(z)| = |d(x, z) - d(y, z)| \leq d(x, y).$$

Thus $\|\Phi(x) - \Phi(y)\|_\infty \leq d(x, y)$. Furthermore note that $|\Phi(x)(x) - \Phi(y)(x)| = |d(x_0, x) - d(y, x) - d(x_0, x)| = d(x, y)$. Thus $\|\Phi(x) - \Phi(y)\| = d(x, y)$. Thus Φ is a isometry and hence a embedding as well.

Problem 3

Let $f \in C^2(\mathbb{R}; \mathbb{R})$ be such that

$$f(x) \geq 0 \quad \text{and} \quad f''(x) \leq 0, \quad \text{for all } x \in \mathbb{R}.$$

Show that f is constant.

Solution. We will show that $\forall x \in \mathbb{R}, f'(x) = 0$. Fix a $x \in \mathbb{R}$. By Taylor's formula we know that

$$f(y) = f(x) + f'(x)(y - x) - \frac{f''(\xi)}{2}(y - x)^2$$

For all $y \in \mathbb{R}$ and some $\xi \in [x, y] \cup [y, x]$. Note that since $f''(\xi) \leq 0$. Thus we have that

$$0 \leq f(y) \leq f(x) + f'(x)(y - x) = L_x(y).$$

Assume the contrary that $f'(x) \neq 0$:

1. $f'(x) > 0$: For

$$y < -\frac{f(x)}{f'(x)} + x$$

we have that

$$L_x(y) < 0 \leq f(x) \leq L_x(y)$$

which is a contradiction.

2. $f'(x) < 0$: For

$$y > -\frac{f(x)}{f'(x)} + x$$

we have

$$L_x(y) < 0 \leq f(x) \leq L_x(y)$$

which is a contradiction. Hence $f'(x) = 0$. This is true for any arbitrary x , hence f is constant

Problem 4 : Basic Exam, Fall 2022

Let (x_n) be a sequence of distinct points in $[0, 1]$ and let (y_n) be a sequence in $\mathbb{R}$ such that

$$|y_j - y_k| \leq 10|x_j - x_k|, \quad \forall j, k \in \mathbb{N}.$$

(a) Show that for each $n \in \mathbb{N}$, there is a function $f_n : [0, 1] \rightarrow \mathbb{R}$ such that

- $f_n(x_j) = y_j$, $1 \leq j \leq n$,
- $|f_n(x) - f_n(x')| \leq 10|x - x'|$, for all $x, x' \in [0, 1]$.

(b) Show that there is a function $f : [0, 1] \rightarrow \mathbb{R}$ such that

- $f(x_j) = y_j$, $\forall j \in \mathbb{N}$,
- $|f(x) - f(x')| \leq 10|x - x'|$, for all $x, x' \in [0, 1]$.

Solution. (a): We will interpolate with peicewise linear functions. Let $a_1, \dots, a_n$ be $x_1, \dots, x_n$ in sorted order, furthermore apply the same rearrangement to $y_1, \dots, y_n$ to get $b_1, \dots, b_n$. Define f_n as follows:

$$f_n(x) := \begin{cases} \frac{b_{i+1}-b_i}{a_{i+1}-a_i}(x-a_i) + b_i & \text{if } x \in [a_i, a_{i+1}] \\ b_1 & \text{if } x \in [0, a_1] \\ b_n & \text{if } x \in [a_n, 1] \end{cases}.$$

We will prove the Lipshitz constant using 4 cases:

1. Let $I = [0, a_1]$ or $[a_n, 1]$: Suppose $x, y \in I$. Then

$$f(x) - f(y) = 0 \leq 10|x - y|.$$

2. Let $I = [0, a_1]$. Suppose $x \in I$ and $y \in [a_i, a_{i+1}]$ ($1 \leq i \leq n-1$).

$$\begin{aligned} |f_n(x) - f_n(y)| &\leq |f_n(x) - f_n(a_1)| + |f_n(a_1) - f_n(a_i)| + |f(a_i) - f(y)| \\ &\leq 10|a_1 - a_i| + \left| \frac{b_{i+1}-b_i}{a_{i+1}-a_i}(y-a_i) \right| \\ &\leq 10|a_1 - a_i| + 10|y - a_i| = 10|a_i - a_1| + |y - a_i| \\ &= 10|a_1 - y| \leq 10|x - y| \end{aligned}$$

Proof is symmetric for $x \in [a_n, 1]$.

3. $x \in [a_i, a_{i+1}]$ and $y \in [a_j, a_{j+1}]$:

$$\begin{aligned} |f_n(x) - f_n(y)| &\leq |f_n(x) - f_n(a_i)| + |f_n(a_i) - f_n(a_j)| + |f_n(a_j) - f_n(y)| \\ &= \left| \frac{b_{i+1}-b_i}{a_{i+1}-a_i}(x-a_i) \right| + 10|a_i - a_j| + \left| \frac{b_{j+1}-b_j}{a_{j+1}-a_j}(y-a_j) \right| \\ &\leq 10|x - a_i| + 10|a_i - a_j| + 10|y - a_j| = 10|x - y| \end{aligned}$$

4. $x \in [0, a_1]$ and $y \in [a_n, 1]$:

$$|f_n(x) - f_n(y)| = |b_1 - b_n| \leq 10|a_1 - a_n| \leq 10|x - y|$$

(b): We will first prove that for all $n \in \mathbb{N}$ that $\|f_n\|_\infty \leq M$ for some $M \in \mathbb{R}$. Let

$$m_n := \max_{1 \leq j \leq n} |y_j|.$$

1. If $x \in [0, a_1]$: $f_n(x) = b_1 \leq m_n$.

2. $x \in [a_n, 1]$: $f_n(x) = b_n \leq m_n$.

3. $x \in [a_i, a_{i+1}]$: $f_{n(x)} \leq \max(b_i, b_{i+1}) \leq m_n$

Thus $f_n(x) \leq m_n$. Now we just need to get a uniform bound for m_n . Note that for all $j \in \mathbb{N}$ we have that

$$|y_j| \leq |y_j - y_i| + |y_1| \leq 10|x_j - x_1| + |y_1| \leq 10 + |y_1|.$$

Thus $M = 10 + |y_1|$ and $m_n \leq M$ meaning $\|f_n\|_\infty \leq M$. Note that $\forall \varepsilon > 0$, let $\delta = \frac{\varepsilon}{10}$. $\forall x, y \in [0, 1]$ where $|x - y| < \delta$ we have that for all $n \in \mathbb{N}$: $|f_n(x) - f_n(y)| \leq 10|x - y| < \varepsilon$. Thus the sequence is equicontinuous. By Arzela Ascoli, there exists a subsequence $(f_{n_k})_{k=1}^\infty$ which converges uniformly to a function f . We have to prove that the two properties hold for f . $\forall m \in \mathbb{N}$, there exists a $K \in \mathbb{N}$ such that for all $k \geq K$, $n_k \geq m$. By the definition of f_n , $f_{n_k}(x_m) = y_m$. Thus the equality holds for f . Since $\forall n \in \mathbb{N}$, $|f_n(x) - f_n(x')| \leq 10|x - x'|$ we also have the inequality for f thus $|f(x) - f(x')| \leq 10|x - x'|$.

Problem 5

Let $\varepsilon > 0$. Show that there exists a constant $C = C_\varepsilon > 0$ such that for all $f \in C^2([- \varepsilon, \varepsilon])$, we have

$$|f'(0)| \leq C\|f\|^{1/2} \left(\|f''\|^{1/2} + \|f\|^{1/2} \right).$$

Here

$$\|f\| = \sup_{|x| \leq \varepsilon} |f(x)|, \quad \|f''\| = \sup_{|x| \leq \varepsilon} |f''(x)|.$$

Solution. $\forall x \in [-\varepsilon, \varepsilon]$ there exists a $\xi \in [0, x] \cup [x, 0]$ where

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2.$$

Thus we have

$$\begin{aligned} -f'(0)x &= f(0) - f(x) + \frac{1}{2}f''(0)x^2 \\ |f'(0)| &\leq \frac{1}{|x|}|f(0)| + \frac{1}{|x|}|f(x)| + \frac{1}{2}|f''(0)x| \\ &\leq 2\frac{\|f\|}{|x|} + \frac{1}{2}|x|\|f''\| \end{aligned}$$

for $|x| \neq 0$.

Suppose $\|f''\| = 0$, then $|f'(0)| \leq \frac{2}{|x|}\|f\|$ for all $x \in [-\varepsilon, \varepsilon] - \{0\}$, Thus $|f'(0)| \leq \frac{2}{\varepsilon}\|f\|$ and $C_1 = \frac{2}{\varepsilon}$. If $\|f''\| > 0$. Consider the function $g : (0, \infty) \rightarrow \infty$ where

$$\begin{aligned}
g(t) &= \frac{2}{t}\|f\| + \frac{1}{2}t\|f''\| \\
g'(t) &= -\frac{2}{t^2}\|f\| + \frac{1}{2}\|f''\| = 0 \\
\frac{1}{2}\|f''\| &= \frac{2}{t^2}\|f\| \\
t^2 &= \frac{4\|f\|}{\|f''\|} \\
t &= \frac{2\|f\|^{\frac{1}{2}}}{\|f''\|^{\frac{1}{2}}} \\
g\left(\frac{2(\|f\|)^{\frac{1}{2}}}{\|f''\|^{\frac{1}{2}}}\right) &= \|f''\|^{\frac{1}{2}}\|f\|^{\frac{1}{2}} + \|f\|^{\frac{1}{2}}\|f''\|^{\frac{1}{2}} = 2\|f\|^{\frac{1}{2}}\|f''\|^{\frac{1}{2}}
\end{aligned}$$

Thus if

$$\frac{2\|f\|^{\frac{1}{2}}}{\|f''\|^{\frac{1}{2}}} \leq \varepsilon$$

we have that

$$|f'(0)| \leq 2(\|f\|\|f''\|)^{\frac{1}{2}} \leq 2\|f\|^{\frac{1}{2}}(\|f\|^{\frac{1}{2}} + \|f''\|^{\frac{1}{2}}).$$

If

$$\frac{2\|f\|^{\frac{1}{2}}}{\|f''\|^{\frac{1}{2}}} > \varepsilon \Rightarrow \|f''\|^{\frac{1}{2}} < \frac{2}{\varepsilon}\|f\|^{\frac{1}{2}},$$

note that g is a decreasing function from $(0, t)$. Thus consider we have that

$$\begin{aligned}
|f'(0)| &\leq 2\varepsilon^{-1}\|f\| + \frac{\varepsilon}{2}\|f''\| \\
&\leq 2\varepsilon^{-1}\|f\| + 2\varepsilon^{-1}\|f\|^{\frac{1}{2}}\|f''\|^{\frac{1}{2}} \\
&\leq 2\varepsilon^{-1}\|f\|^{\frac{1}{2}}(\|f\|^{\frac{1}{2}} + \|f''\|^{\frac{1}{2}})
\end{aligned}$$

Let $C_\varepsilon = \max(2\varepsilon^{-1}, 2)$. Thus

$$|f'(0)| \leq C_\varepsilon\|f\|^{\frac{1}{2}}(\|f\|^{\frac{1}{2}} + \|f''\|^{\frac{1}{2}})$$

Problem 6 : Basic Exam, Fall 2015

Let $X = \mathbb{R} \setminus \{0\}$. Find a metric ρ on X with the following properties:

- (X, ρ) is a complete metric space.
- If $\{x_n\}_{n=1}^\infty$ is a sequence in X and $x \in X$, then

$$\lim_{n \rightarrow \infty} |x_n - x| = 0 \iff x_n \rightarrow x \quad \text{in } (X, \rho).$$

Prove both properties, as well as all of your other assertions, in full detail.

Solution. We define our metric as follows:

$$\rho(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right| + |x - y|$$

- **Nonnegativity and Symmetry:** This is inherited from the absolute value.
- **$\rho(x, x) = 0$:** $\left| \frac{1}{x} - \frac{1}{x} \right| + |x - x| = 0$.
- **Triangle Inequality:**

$$\begin{aligned} \rho(x, y) &= \left| \frac{1}{x} - \frac{1}{y} \right| + |x - y| \\ &= \left| \frac{1}{x} - \frac{1}{y} - \frac{1}{z} + \frac{1}{z} \right| + |x - y - z + z| \\ &\leq \left| \frac{1}{x} - \frac{1}{z} \right| + \left| \frac{1}{y} - \frac{1}{z} \right| + |x - z| + |y - z| \\ &= \rho(x, z) + \rho(z, y) \end{aligned}$$

Suppose $(x_n)_{n=0}^{\infty}$ is Cauchy in (X, ρ) . Thus $\forall \varepsilon > 0$, there exists a $N \geq 0$ such that

$$\begin{aligned} \varepsilon &> \left| \frac{1}{x_m} - \frac{1}{x_n} \right| + |x_m - x_n| \\ &= \frac{|x_n - x_m|}{|x_m x_n|} + |x_m x_n|(|x_n - x_m|) \\ &= \frac{(1 + |x_m x_n|)|x_n - x_m|}{|x_m x_n|} > |x_m - x_n| \end{aligned}$$

Thus (x_n) Cauchy in $(\mathbb{R}, |\cdot|)$. Let x be the limit in $(\mathbb{R}, |\cdot|)$. There exists a $N \in \mathbb{N}$ such that $\forall m \geq N$ $|x_m - x| < \varepsilon$.

$$\begin{aligned} \rho(x_m, x) &= \left| \frac{1}{x_m} - \frac{1}{x} \right| + |x_m - x| = \frac{(1 + |x_m x|)|x_m - x|}{|x_m x|} \\ &\leq |x_m - x| < \varepsilon \end{aligned}$$

Thus $x_m \rightarrow x$. Hence (X, ρ) is complete.

Suppose

$$\lim_{n \rightarrow \infty} |x_n - x| = 0$$

for (x_n) in X and $x \in X$. By the above calculation $x_n \rightarrow x$ in (X, ρ) . Suppose $x_n \rightarrow x$ in (X, ρ) . Then $\forall \varepsilon > 0$ there exists a $N \in \mathbb{N}$ such that $\rho(x_n, x) < \varepsilon$ for all $n \geq N$. Thus

$$|x_n - x| \leq \frac{(|x_n x| + 1)|x_n - x|}{|x_n x|} = \rho(x_n, x) < \varepsilon.$$

Thus

$$\lim_{n \rightarrow \infty} |x_n - x| = 0$$

Problem 7 : cf. Basic Exam, Fall 2025

Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable on the compact interval $[a, b]$ (with one-sided derivatives at the end points), such that $f'(a) \neq f'(b)$. Show that $f'(x)$ assumes all values between $f'(a)$ and $f'(b)$ in the interval $a < x < b$.

Hint: Consider the function $x \mapsto f(x) - \mu x$ for a suitable μ .

Solution. Without loss of generality, $f'(a) < f'(b)$. Suppose $\mu \in [f'(a), f'(b)]$. Consider $g(x) = f(x) - \mu x$. Then g is also differentiable on $[a, b]$ and $g'(x) = f'(x) - \mu$. Note that $g'(a) < 0$ and $g'(b) > 0$. g is a continuous function on a compact set so it must achieve its extreme value. Thus the point with the extreme value must be in the interior or at the end points. Suppose $\exists x$ such that $g(x) \geq g(y)$ for all $y \in [a, b]$: $g'(x) = 0 \Rightarrow f'(x) = \mu$, and we are done.

Endpoints: Suppose $g(a) \leq g(y)$ for all $y \in [a, b]$. By assumption $g'(a) < 0$, thus $\exists \delta > 0$ such that for all $y \in [a, a + \delta)$:

$$\left| \frac{g(y) - g(a)}{y - a} - g'(a) \right| < |g'(a)| \Rightarrow \frac{g(y) - g(a)}{y - a} < 0 \Rightarrow g(y) < g(a)$$

Thus we have a contradiction and the minimum cannot be achieved at $x = a$. By a similar argument the minimum cannot be achieved at $x = b$. Thus the maximum of g is achieved at $x = a$ and $x = b$. Thus $\forall x \in (a, b)$, $g(x) \leq g(a) = g(b)$. Look at the function $h(x) = g(x) - g(a)$. h is differentiable on (a, b) and continuous on $[a, b]$ and $h(a) = 0 = h(b)$. By Rolle's Theorem, $\exists c \in (a, b)$ such that $h'(c) = 0 \Rightarrow g'(c) = 0$. Thus $f'(c) = \mu$.